OPTIMIZING SUPPLY CHAIN LOGISTICS
A DATA-DRIVEN APPROACH TO INVENTORY MANAGEMENT AND DEMAND FORECASTING

Project Report Submitted in fulfillment of the Requirements for the Data Science & AI Training Program by IDRA

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DECLARATION

I declare that this capstone report represents my own analytical work completed as part of the IDRA Data Science & AI Training Program. The analysis and code were prepared for this project, relevant materials have been acknowledged, and I understand the methods and results presented.

ACKNOWLEDGEMENT

I thank the India Data Research Academy (IDRA), the training faculty and my supervisor for the guidance and learning opportunities provided through the Data Science & AI Training Program. I also acknowledge the project dataset and supporting materials supplied through the LMS.

ABSTRACT

This study investigates supply chain demand patterns and develops a predictive model for daily Units_Sold using the Project 3 dataset provided through the IDRA LMS. The dataset contains 91,250 observations and 15 original variables spanning 2024-01-01 to 2024-12-30. Data-quality checks found no missing values and no duplicate records. Date was converted to datetime format, while calendar, inventory-gap and price-margin features were engineered. Demand_Forecast was strongly associated with actual Units_Sold (r = 0.949), and promoted observations had higher average demand (24.91 versus 19.50 units). A chronological train-test split evaluated later observations. A Ridge regression pipeline combined numerical scaling with one-hot encoding. On unseen data, MAE = 2.32, RMSE = 2.92, and R^2 = 0.768. The findings demonstrate useful predictive signal for inventory planning while highlighting the need for leakage checks and additional temporal validation before deployment.

Keywords: supply chain analytics; demand forecasting; inventory management; EDA; regression; predictive analytics.

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1. INTRODUCTION

Supply chains must balance product availability against inventory cost. Reliable demand estimates can support replenishment planning, warehouse allocation and purchasing decisions. This project applies a complete data science workflow to operational supply-chain data to understand demand behaviour and predict daily Units_Sold.

2. PROBLEM STATEMENT AND MOTIVATION

How accurately can daily Units_Sold be predicted using product, warehouse, supplier, regional, inventory, pricing, promotion and demand-forecast information in the supplied dataset? A reliable data-driven estimate can support better inventory decisions.

3. OBJECTIVES

• Understand dataset structure and data quality.

• Clean and validate the dataset.

• Identify demand patterns through EDA and statistics.

• Engineer calendar, inventory and price features.

• Build a regression model to predict Units_Sold.

• Evaluate later-period generalisation and provide evidence-based recommendations.

4. RESEARCH QUESTIONS

• What are the dataset's main characteristics and quality issues?

• How does demand vary across promotion status and regions?

• Which variables are most strongly associated with Units_Sold?

• How accurately can the model predict later observations?

• What recommendations follow from the findings?

5. DATASET DESCRIPTION AND DATA UNDERSTANDING

The file P_3_supply_chain_dataset1.csv was provided through the IDRA LMS for approved Project 3. The guide classifies this topic as a supply-chain regression/forecasting project with demand or sales as the intended target.

Item	Result
Observations	91,250
Original variables	15
Period	2024-01-01 to 2024-12-30
Target	Units_Sold
SKUs	50
Warehouses	5
Suppliers	10
Regions	4

Table 1. Summary of the dataset.

5.3 Data Dictionary

Variable	Meaning	Type	Role
Date	Observation date	object	Feature/context
SKU_ID	Product identifier	object	Feature/context
Warehouse_ID	Warehouse identifier	object	Feature/context
Supplier_ID	Supplier identifier	object	Feature/context
Region	Geographic region	object	Feature/context
Units_Sold	Actual units sold	int64	Target
Inventory_Level	Available inventory	int64	Feature/context
Supplier_Lead_Time_Days	Supplier lead time	int64	Feature/context
Reorder_Point	Reorder threshold	int64	Feature/context
Order_Quantity	Quantity ordered	int64	Feature/context
Unit_Cost	Cost per unit	float64	Feature/context
Unit_Price	Selling price	float64	Feature/context
Promotion_Flag	Promotion indicator	int64	Feature/context
Stockout_Flag	Stockout indicator	int64	Feature/context
Demand_Forecast	Forecasted demand	float64	Feature/context

6. DATA CLEANING

All original columns contained zero missing values, and exact duplicate checking returned zero duplicate rows. Date was converted from object to datetime. Identifier variables remained categorical. IQR diagnostics were used to screen numerical extremes, but observations were not automatically deleted because unusual operational values may be meaningful. No rows were removed solely because of outlier status.

Check	Before	Action	After
Missing values	0	No imputation needed	0
Duplicate rows	0	None removed	0
Date type	object	Converted	datetime64
Rows	91250	No removal	91250

Table 3. Data quality before and after cleaning.

7. DATA PREPROCESSING

SKU_ID, Warehouse_ID, Supplier_ID and Region were one-hot encoded. Numerical predictors were standardised inside a Scikit-learn pipeline fitted on training data only. A chronological split used observations through 18 October 2024 for training and later observations for testing. Stockout_Flag was excluded because it was constant. The final feature set also included inventory, lead time, reorder point, order quantity, costs, price, promotion, demand forecast, calendar features, Inventory_Gap, Price_Margin and Markup_Ratio.

8. EXPLORATORY DATA ANALYSIS

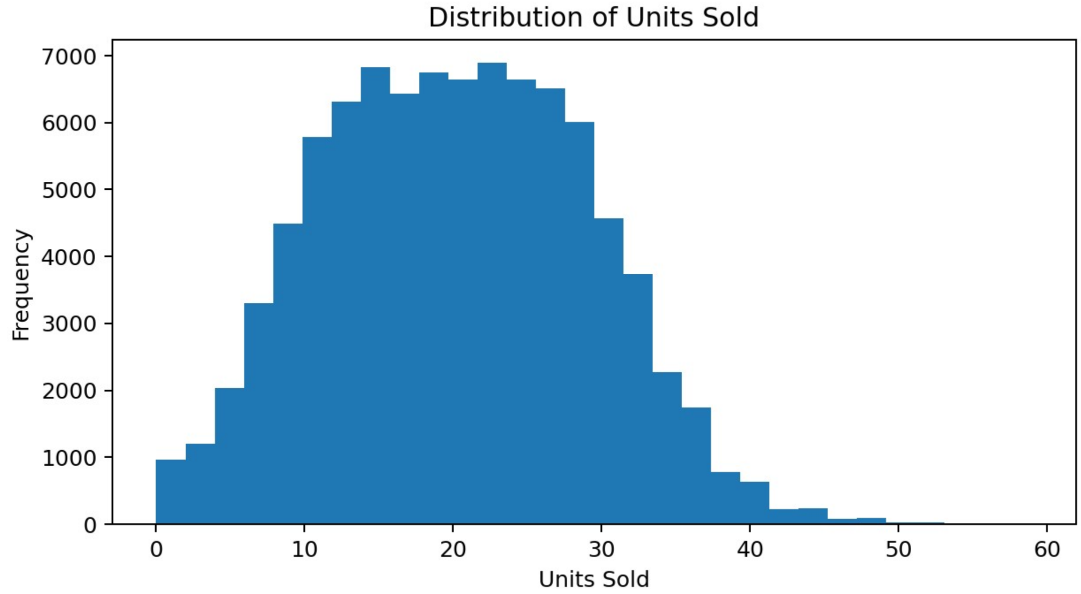

Figure 1. Distribution of Units Sold.

Units_Sold had mean 20.05, median 20 and standard deviation 9.07.

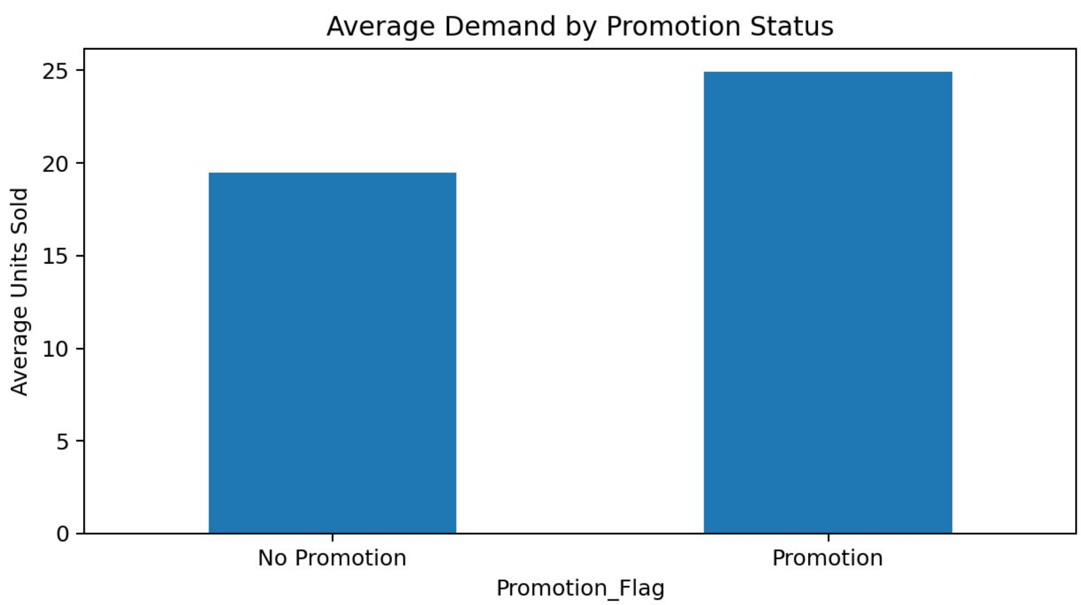

Figure 2. Average demand by promotion status.

Promoted observations averaged 24.91 units versus 19.50 without promotion. This is association rather than causal proof.

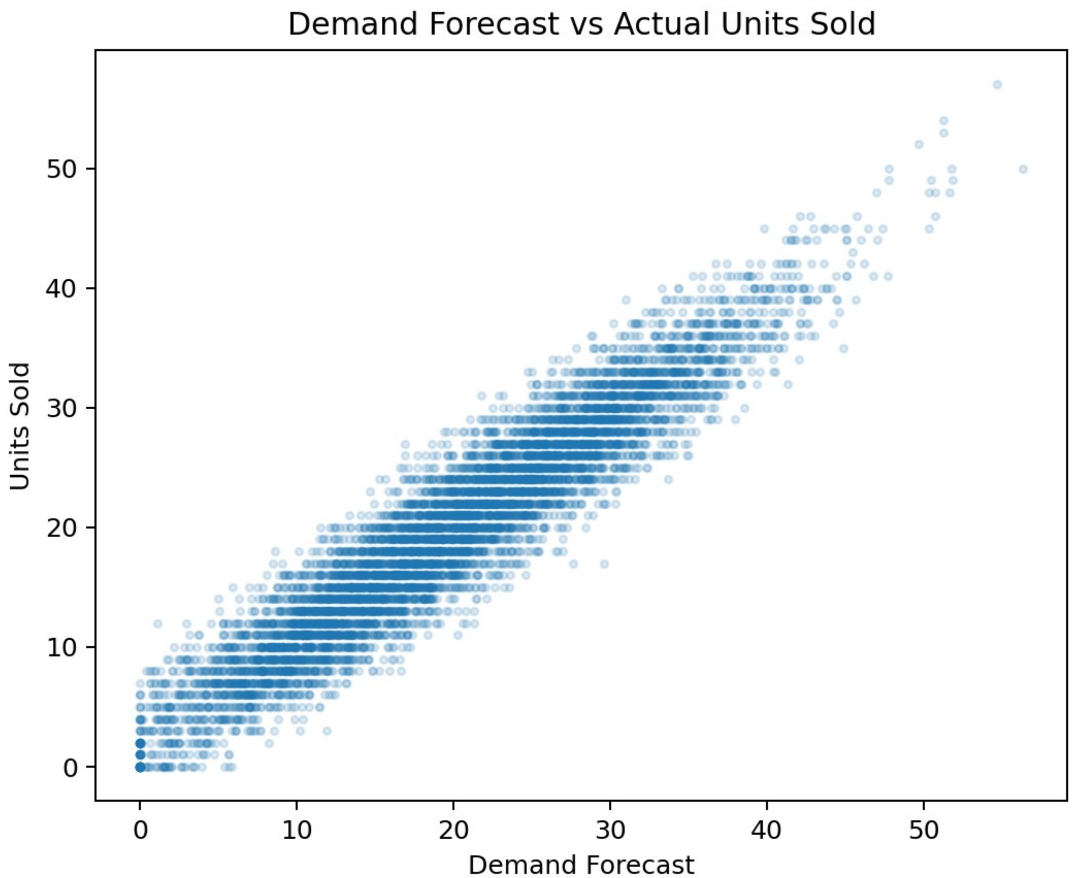

Figure 3. Demand Forecast versus Actual Units Sold.

Demand_Forecast had the strongest numerical relationship with the target (r = 0.949). Its availability at prediction time must be verified in any real deployment.

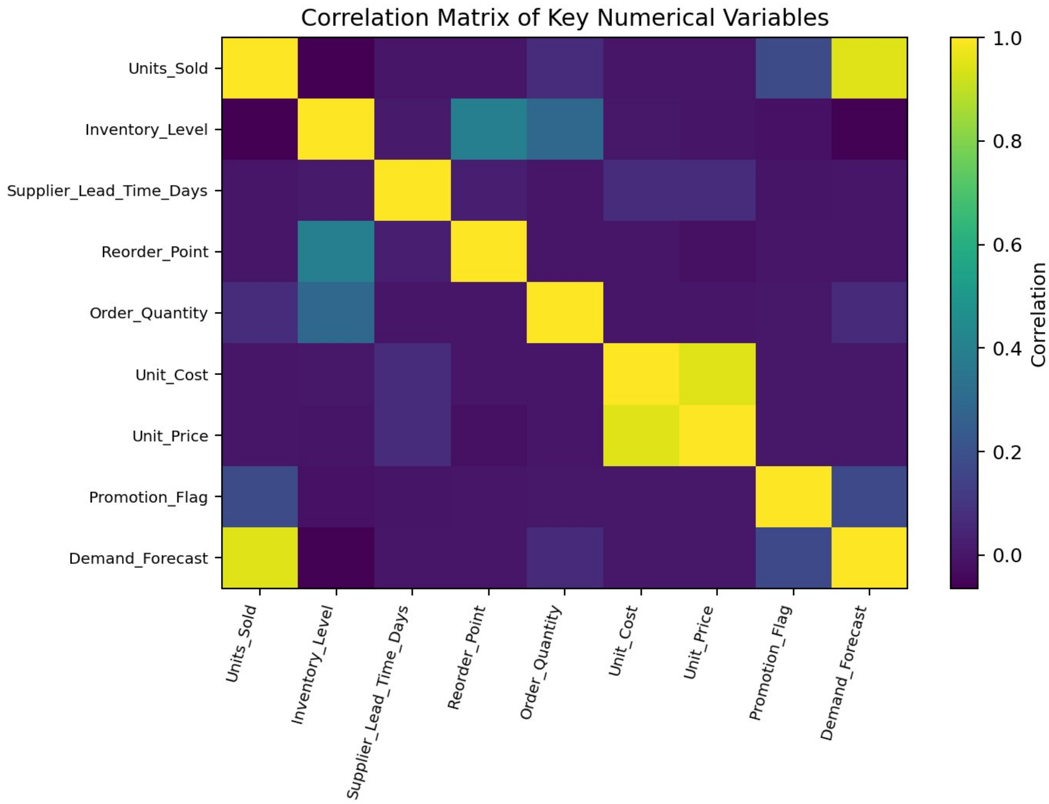

Figure 4. Correlation Matrix of Key Numerical Variables.

The heatmap summarises linear relationships and was used to support feature selection rather than causal claims.

9. STATISTICAL ANALYSIS

Variable	mean	median	std	var	min	max
Units_Sold	20.055	20.0	9.069	82.24	0.0	59.0
Inventory_Level	471.522	461.0	133.488	17819.047	168.0	990.0
Order_Quantity	19.272	0.0	82.341	6780.012	0.0	499.0
Demand_Forecast	20.082	19.95	9.504	90.325	0.0	61.42

Table 4. Descriptive statistics. Regional mean demand ranged from 20.02 to 20.11 units, while promoted observations averaged 5.41 more units than non-promoted observations.

10. FEATURE ENGINEERING

Feature	Construction	Purpose
Month	Date month	Calendar pattern
DayOfWeek	Date weekday	Weekly pattern
Quarter	Date quarter	Seasonal period
Inventory_Gap	Inventory - Reorder point	Inventory position
Price_Margin	Price - Cost	Unit margin
Markup_Ratio	Margin / Cost	Relative margin

11. MODEL DEVELOPMENT

The task is supervised regression. Ridge regression was selected as an interpretable regularised baseline suitable for a one-hot encoded feature space. A Pipeline applied StandardScaler to numerical features and OneHotEncoder to categorical features before fitting Ridge(alpha=1.0).

12. MODEL EVALUATION

Metric	Training	Testing
MAE	2.182	2.319
MSE	7.473	8.511
RMSE	2.734	2.917
R^2	0.911	0.768

Table 6. Regression performance. On unseen data, average absolute error was approximately 2.32 units. Training R^2 was 0.911 and test R^2 was 0.768; the gap indicates some generalisation loss but substantial held-out performance.

13. RESULTS AND FINDINGS

• Demand_Forecast was the strongest numerical correlate ($r = 0.949$).

• Promoted observations averaged 24.91 units compared with 19.50 without promotions.

• Regional average demand was comparatively stable.

• The final model achieved test MAE 2.32, RMSE 2.92 and R^2 0.768.

• Chronological testing was harder than training, reinforcing the importance of future-like evaluation.

14. DISCUSSION

The supplied demand forecast was the dominant signal, and promotion status was associated with higher demand. In a real application, forecast availability must be confirmed before the sales outcome occurs to prevent target leakage. Stable regional averages suggest that broad geography is less informative than product and operational signals. Later-period performance was lower than training performance, so model monitoring and repeated temporal validation are important.

15. LIMITATIONS

• Dataset provenance and generation details are limited to the LMS material.

• Demand_Forecast requires a formal leakage audit.

• Relationships in this study are not causal.

• The reported model is a regularised linear baseline rather than an exhaustive model comparison.

• One year of data may not capture long-term structural changes.

16. CONCLUSION

This capstone completed the required data science workflow for supply-chain demand prediction. The dataset was clean with respect to missing values and duplicates, and feature engineering added calendar, inventory and pricing context. EDA identified a strong forecast-demand relationship and higher average demand during promotion observations. The chronological Ridge model achieved test MAE 2.32, RMSE 2.92, and R^2 0.768. The project shows how operational data can support evidence-based inventory planning while requiring careful leakage checks and continued validation.

17. RECOMMENDATIONS AND FUTURE WORK

• Verify that demand forecasts are available before actual sales and use them only when leakage-free.

• Plan additional inventory around promotion periods, supported by higher observed average demand.

• Monitor Inventory_Gap alongside reorder thresholds.

• Re-evaluate the model on new time periods.

• Retain operational extremes unless a clear data-quality reason supports removal.

Future work should compare additional models, add lagged demand and rolling statistics, incorporate holidays or external signals, validate across multiple future periods, and implement formal monitoring.

18. REFERENCES

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Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

APPENDIX A. CODE

Complete reproducible code is supplied in Sharma_Pratyush_Capstone_Notebook.ipynb.

APPENDIX B. ADDITIONAL FIGURES AND TABLES

Supporting checks and outputs are available in the notebook. The final cleaned/preprocessed dataset is supplied as Sharma_Pratyush_Capstone_Cleaned_Preprocessed.csv.